What is security on MCU?



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Security

Definition

Security is the degree of resistance to, or **protection** from, harm.

It applies to any vulnerable and valuable **asset**, such as a person, dwelling, community, nation, or organization.

Wikipedia

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Security

Assets

- Security is about protecting assets
- Information, capability, advantage, feature, financial or technical resource that may be damaged, lost or disrupted
  - Consumers Personal Information: Financial, Health, Location, Passwords, Accounts
  - A Product or Solution: Processes, Services, Intellectual Property, Firmware, Brand
  - Health and Safety: Medical devices, Manufacturing Processes and Equipment, Transport & Vehicles
  - The Work Place: Production Equipment, Environmental and Access Controls
- Digital (software sources), physical (a car or a server) or commercial (brand)
- Damage to an asset may affect the normal operation of the system (Denial of Services) as well as that of individuals and organizations involved with the system

Security

CIA security properties

- Properties to be protected depend on applications & assets
- **C**onfidentiality
  - Information is not made available or disclosed to unauthorized individuals, entities, or processes
- **I**ntegrity
  - Maintain and assure the accuracy and completeness of data over its entire life-cycle. Data cannot be modified in an unauthorized or undetected manner.
- **A**vailability
  - Information available when needed. Computing systems used to store and process information, security controls used to protect it, and communication channels used to access it function correctly.



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Leveraging ST's ecosystem to secure your STM32 Application

Smart World

IoT enables intelligent services





Smart Home



Smart City



Smart Industry



Smart Driving



eHealth

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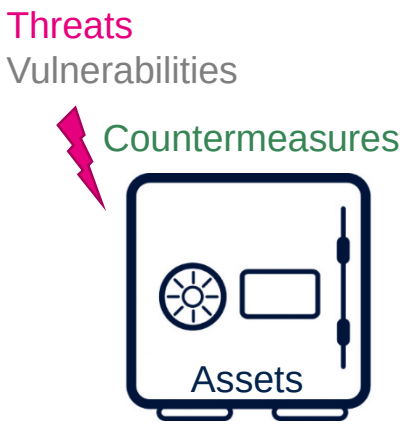
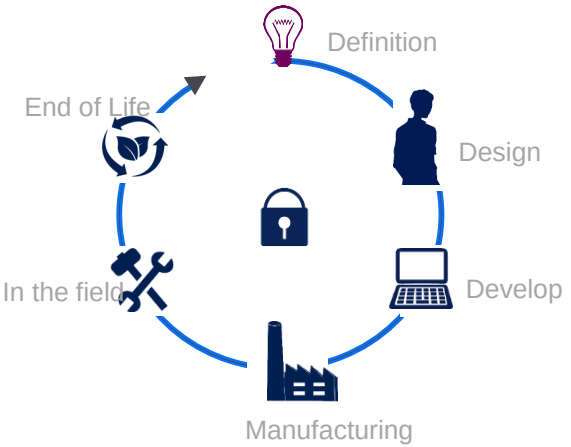
IoT Systems

Main Threats



- Access / misuse of services and networks
- Access / misuse of devices
- Theft of confidential data or identity
- Counterfeit devices or services

Building security in 3 steps

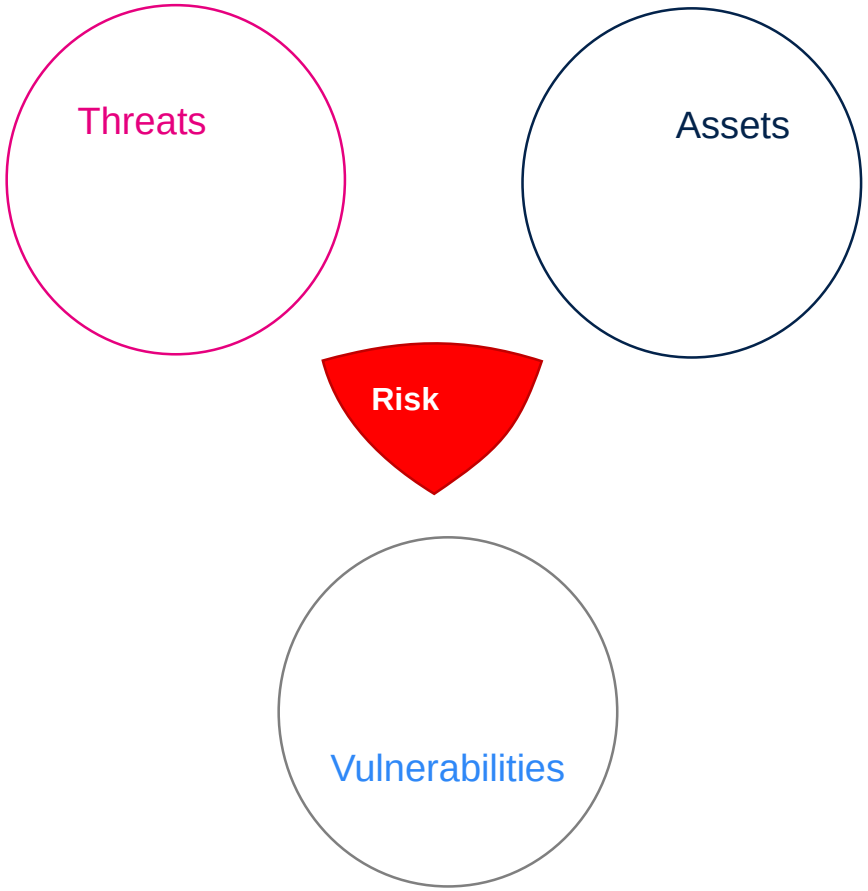
Threat analysis	Security at system level	Customer product Life Cycle
		
Analyze your system	Secure all elements of the system or network	Build secure solution from design to disposal

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Analyze your System

Threat analysis

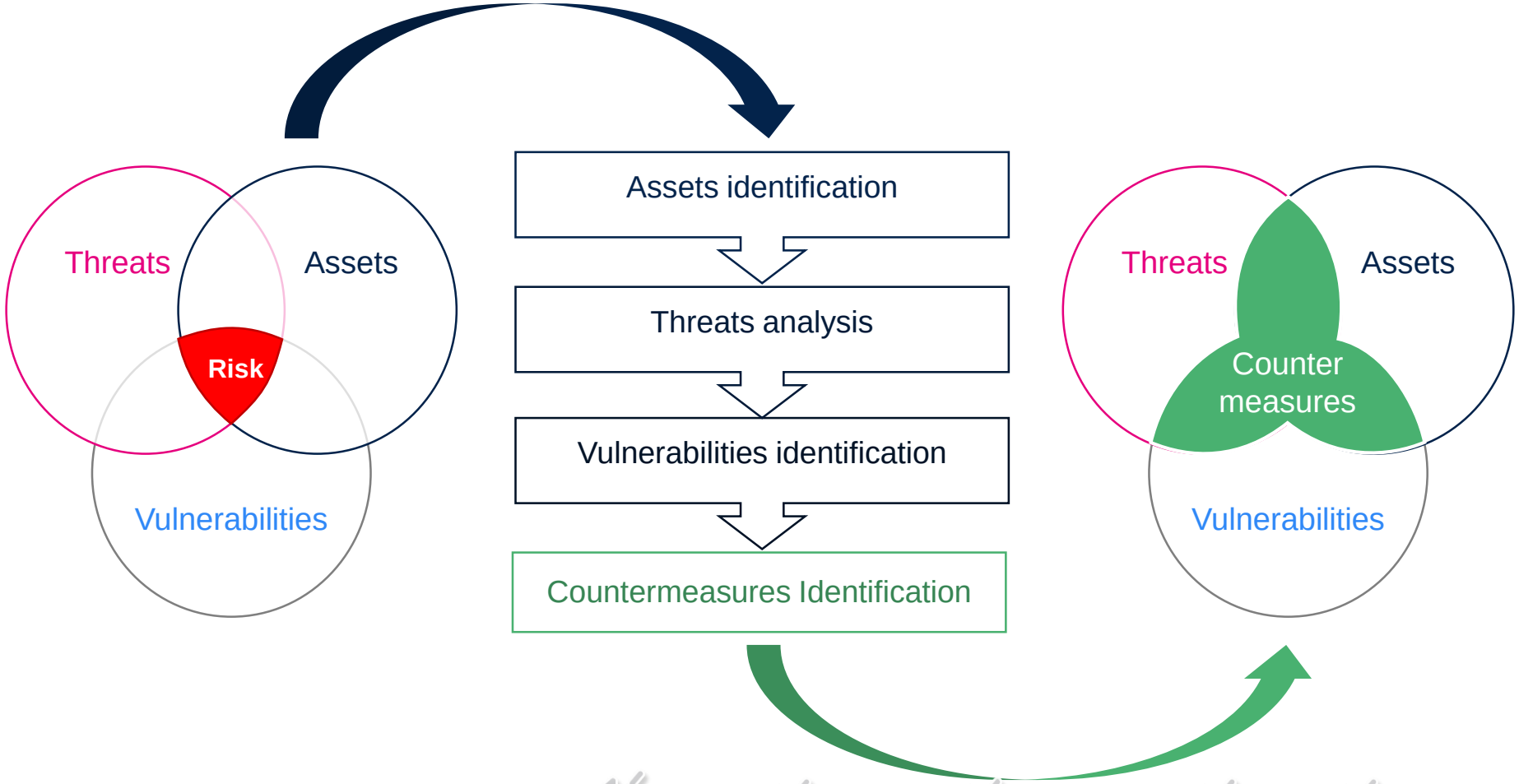
Main definition



- *An Asset is what we want to protect.*
- *A Threat is what we need to protect against.*
- *A Vulnerability is a weakness or gap in our protection efforts.*
- *Risk is the intersection of assets, threats, and vulnerabilities.*

Threat analysis

Follow a clear process/methodology

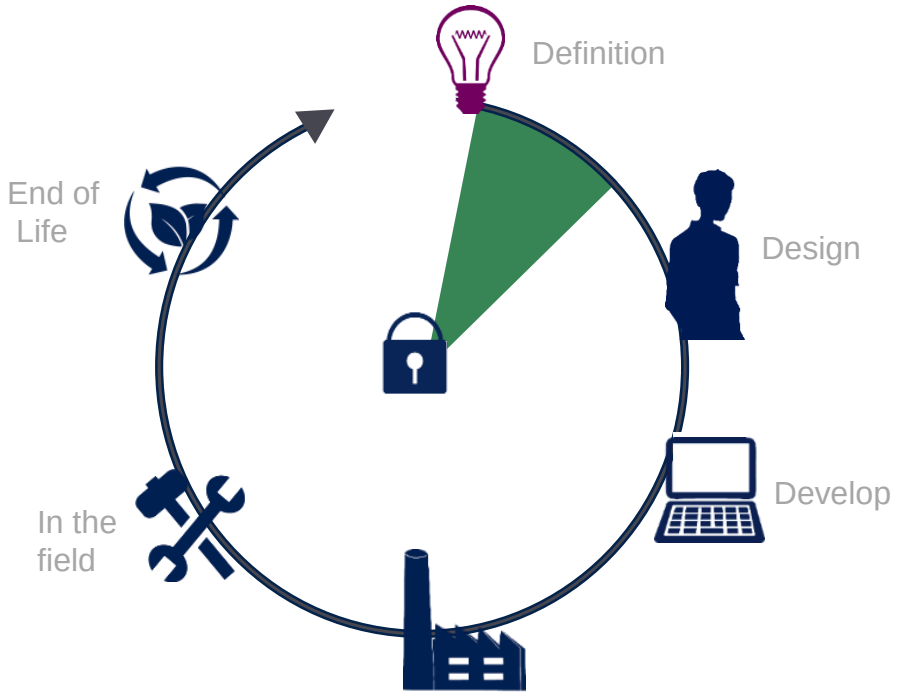


Build secure solution from design to disposal

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Customer product life cycle

Definition and Design : Threat analysis



The context

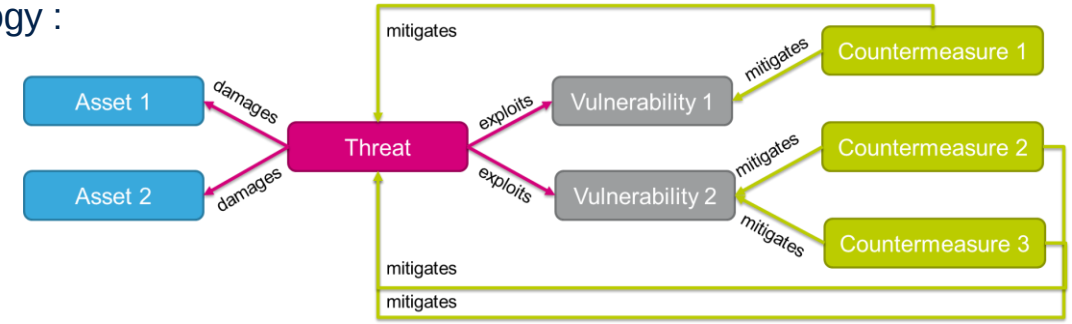


The project team at customer side, need to go through the threat analysis.

Identifying their assets.
Analyzing the threats,
looking for vulnerabilities,
and countermeasures.

To learn more

Methodology :



http://www.ptatechnologies.com/PTA5.htm

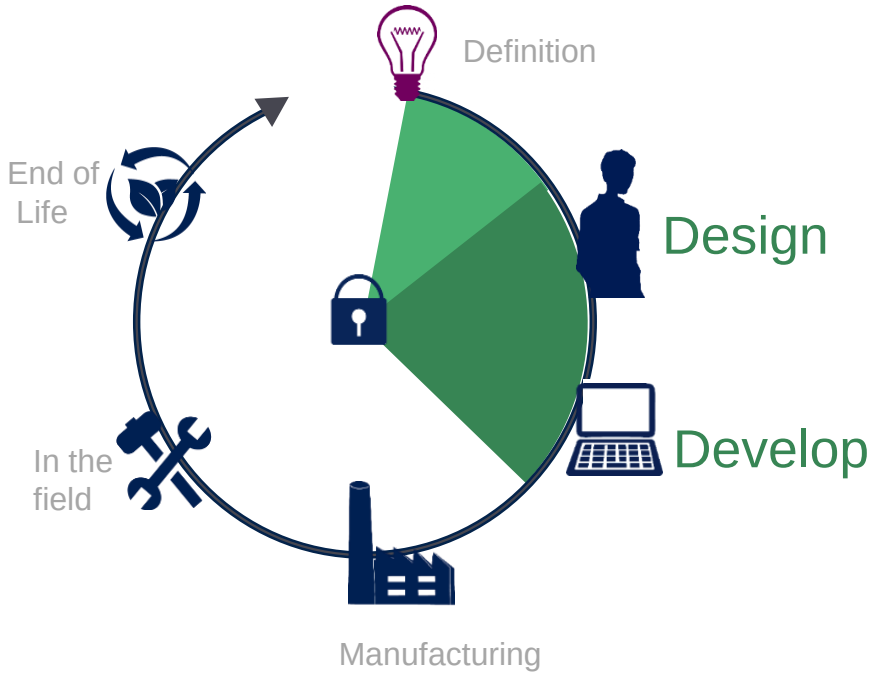
Examples

https://developer.arm.com/support/training/arm-psa-threat-model-analysis

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Customer Product Life Cycle

Design and development: Threat analysis



To learn more

www.st.com

www.st.com/stm32Trust

The context



The project team at customer, need to know what are the hardware solutions, the software libraries, solutions available to help him develop its application.

Solution Proposal

Wide Product portfolio and Strong security ecosystem	Secure Boot	Debug Lock	MPU	Firewall TrustZone Dualcore	Can add a STSafe Secure Element	X-Cube-Cryptolib CAVP certified	X-Cube-SBSFU	ARM PSA TF-M	FastROM Prog. Service	Secure Firmware Install
STM32G0/G4	√	√	√		√	√	√		√	
STM32L0/L4 *SFI on selected P/Ns only	√	√	√	FW	√	√	√		√	√*
STM32F4/F7	√	√	√		√	√	√		√	
STM32H7	√	√	√		√	√	√		√	√
STM32L5	√	√	√	TZ	√	√		√	√	√
STM32WB	√	√	√	Dual	√	√	√		√	

Security solution with STM32 & STSAFE

The right combination of STM32 and STSAFE

Use STM32 security features

offering various security levels linked to FW security development knowhow by customers

- Memory Protection
- Read out protection level
- Write protection
- Execute only PCROP
- Crypto Hardware (AES, PKA, TRNG)
- Memory Hide Protect (SecureBoot)
- TrustZone™
- Backup RAM memory/service continuity

- CAVP certified STM32 Cryptographic library
- Secure Firmware install (SFI)
- Secure Boot Secure Firmware Upgrade (SBSFU)







Add a Certified (CC EAL5+) Secure Element STSAFE for security improvements.

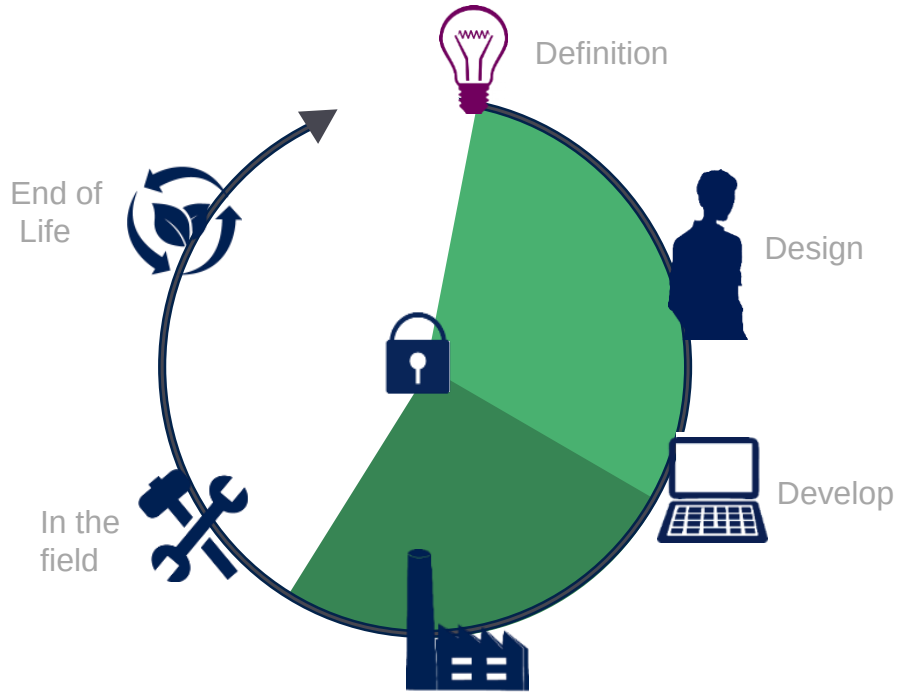
which guaranties optimum security against Remote, Board and silicon level attacks

- Specific countermeasures against silicon level attacks
- Strong authentication / crypto services
- Secure key Storage and handling
- Secure code storage and execution inside SE
- Secure certified key provisioning

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Customer Product Life Cycle

Customer Manufacturing with STM32



Manufacturing

The context



The device owner uses a contract manufacturer to program the device, and wants to protect its code from leaking. As well as prevent overproduction.



Solution Proposal

- To secure the device owner's code, all the way to the final application by decrypting it inside the MCU.
- To count the produced pieces in a trusted way, by using an HSM for example

To learn more

Secure Firmware install
STM32L4,L5,H7 …

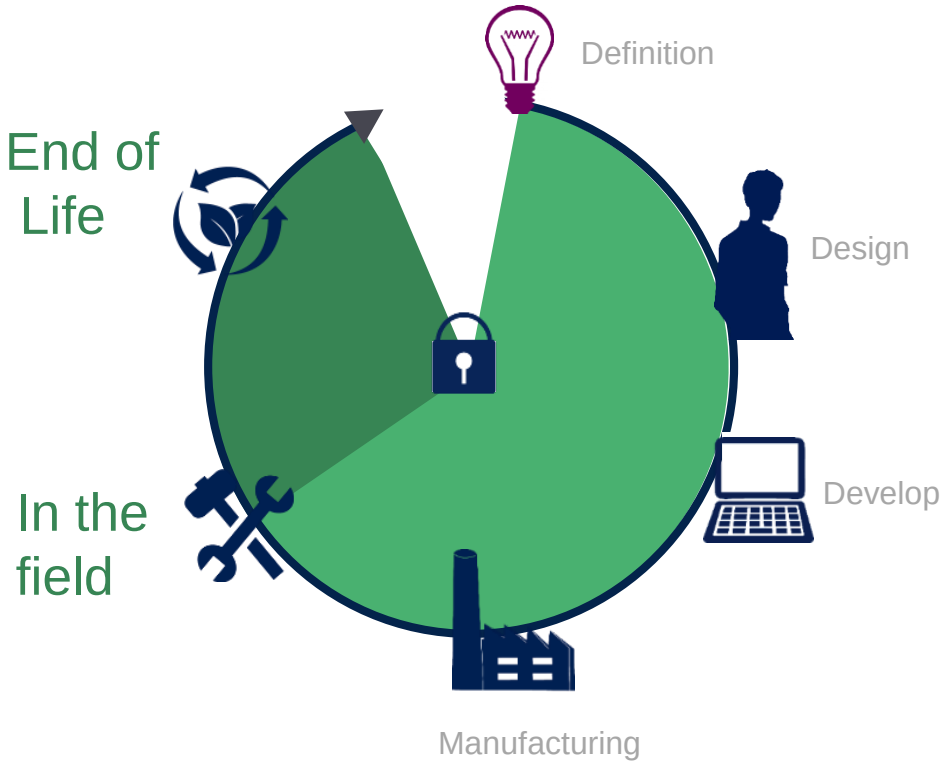
Available documentation :

AN4992	UM2237
AN5054	UM2238

Go to www.st.com/stm32Trust

Customer Product Life Cycle

In the field and product end of life



The context



The **device owner** wants to make sure what is running on his devices is his firmware.

The **device owner** also need to keep the device secured once in the field (secure communication, secure update, decommissioning, etc…)

Solution Proposal

- Implement a root of trust in your application
- Communicate over secured and trusted channels
- Check Firmware update authenticity, integrity…
- …

To learn more

TF-M (on STM32L5)
Trusted Firmware Cortex-M

www.st.com/stm32Trust

Secure Boot,
Secure Firmware Update

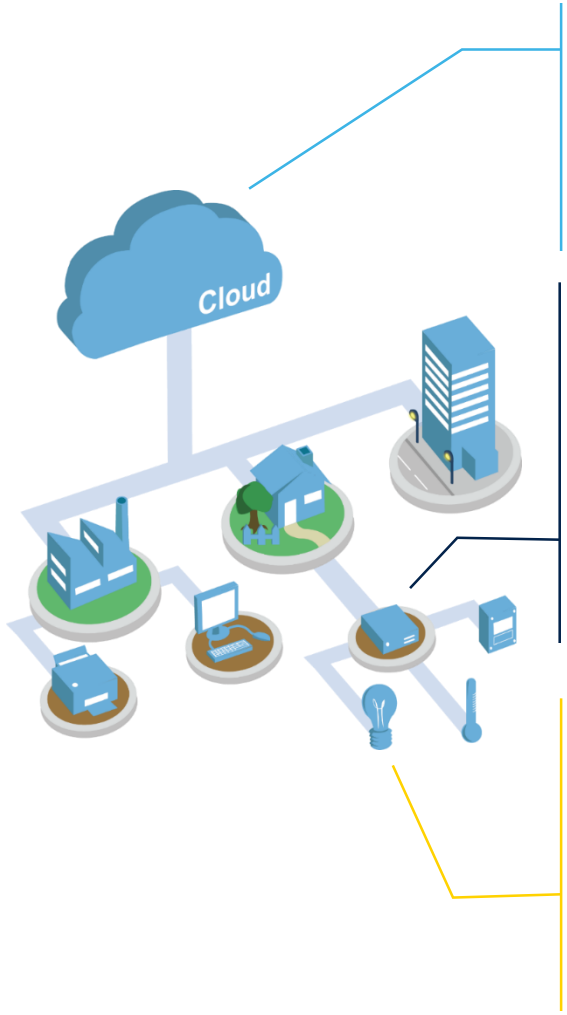
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Secure all elements of the system or network

An example of security at system level

Security at system level



IoT server security :
- Cloud security , User privacy
- Key Management system
- Device Management (diagnostic, monitoring, FW update)
- And other Cloud/Network security strategies

Gateway :
- Key Security node
- Optimized perf. /cost ratio

Proposal:
- Secure manufacturing
- Secure Boot + Secure Firmware update
- Secure communication (dual authentication)
- Secure storage
- …

STM32G4, L4, L5, F4, F7, H7, MP1 and more…

Secure Firmware Install

Secure Boot,
Secure Firmware Update

TF-M (on STM32L5)
Trusted Firmware Cortex-M

Sensor Node :
- Price sensitive device
- Limited resources (RAM,Flash,MHz…)
- Low power device

Proposal :
- Secure Boot Secure Firmware update : making sure your device is running your SW
- Key component isolation using STM32's Firewall
- Product life cycle management via Gateway

STM32G0, G4, L4, L5, WB, and more…

Secure Boot,
Secure Firmware Update

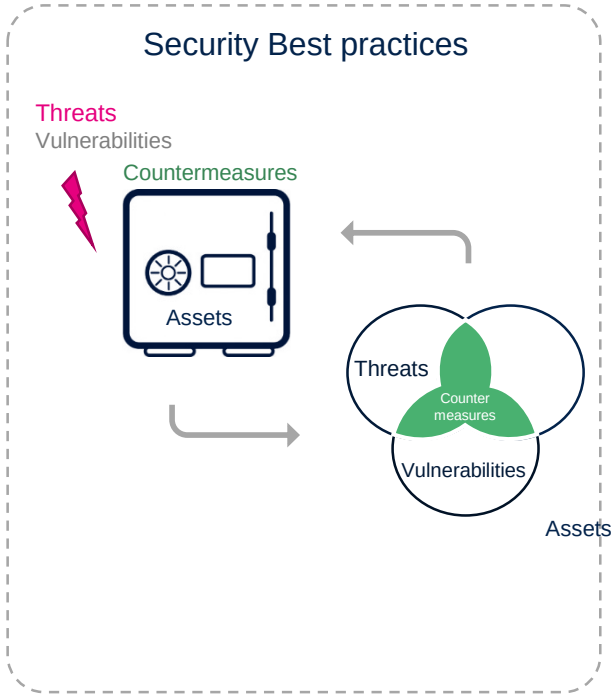
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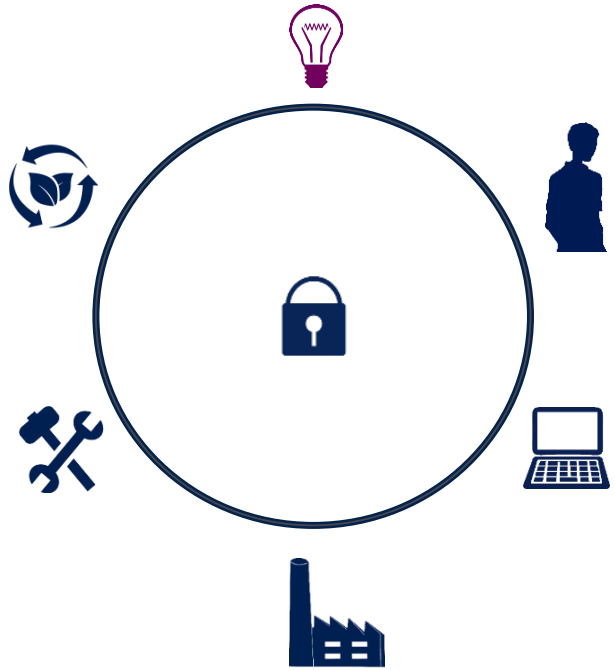
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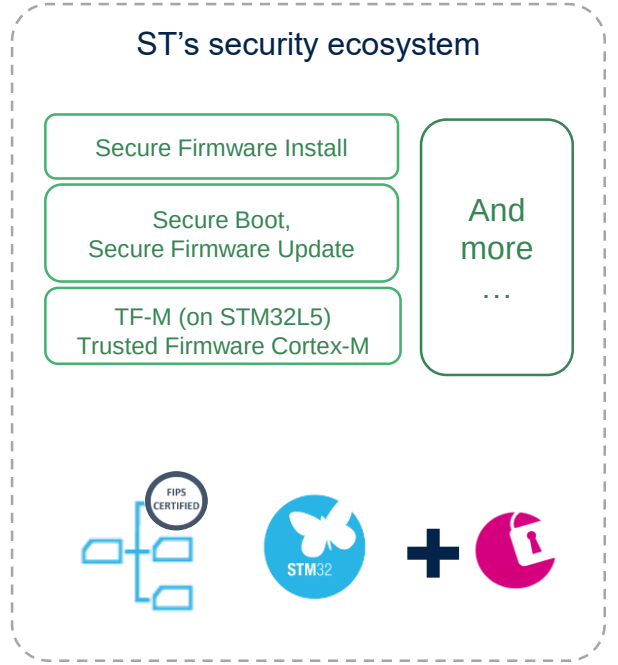
Take Away

Take away

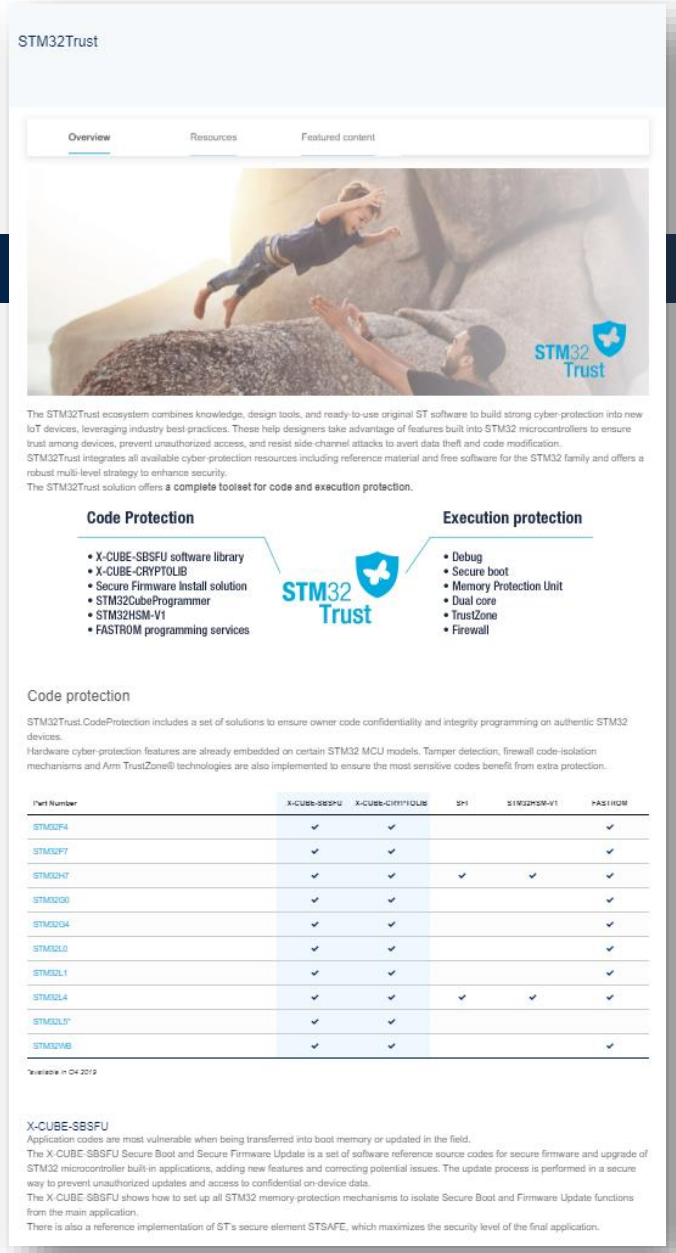
Work with ST, your experienced partner







STM32Trust



Work with ST, your experienced partner

Knowledge Base + Design tools + Ready to use ST solution + Ready for STM32

= STM32 Trust

A new webpage with all STM32's security related resources in one place for easier reference:

www.st.com/stm32trust

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Thank you

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